DESCRIPTION OF COURSEWORK

Course Code	G0123
Course Name	Mathematical Graphics
Lecturer	Goh Chien Yong, Keng Ying Ying, Nurhidayah Binti Hamdan
Academic Session	2025/02
Assessment Title	Assignment 1

A. Introduction/ Situation/ Background Information

Imagine you are designing a digital version of a classic spinner game like the "Wheel of Fortune." This game involves spinning a wheel divided into several sectors, each with a different reward or outcome. In this project, you will use MATLAB to simulate the spinner, visualize the outcomes, and analyse the game's fairness and probabilities.

This assignment focuses on enhancing students' ability to:

1. Use MATLAB's graphical functions to create professional and visually appealing outputs.
2. Reproduce mathematical graphics with precision using MATLAB.
3. Interpret and analyse probabilistic outcomes through simulations.

B. Course Learning Outcomes (CLO) covered

At the end of this assessment, students are able to:

CLO 1 Display ability to use graphic features of software like MATLAB or Scilab.

CLO 2 Reproduce professional mathematical graphics using a software.

C. University Policy on Academic Misconduct

1. Academic misconduct is a serious offense in Xiamen University Malaysia. It can be defined as any of the following:
   i. **Plagiarism** is submitting or presenting someone else's work, words, ideas, data or information as your own intentionally or unintentionally. This includes incorporating published and unpublished material, whether in manuscript, printed or electronic form into your work without acknowledging the source (the person and the work).

ii. **Collusion** is two or more people collaborating on a piece of work (in part or whole) which is intended to be wholly individual and passed it off as own individual work.

iii. **Cheating** is an act of dishonesty or fraud in order to gain an unfair advantage in an assessment. This includes using or attempting to use, or assisting another to use materials that are prohibited or inappropriate, commissioning work from a third party, falsifying data, or breaching any examination rules.

2. All assessments submitted must be the student's own work, without any materials generated by AI tools, including direct copying and pasting of text or paraphrasing. Any form of academic misconduct, including using prohibited materials or inappropriate assistance, is a serious offense and will result in a zero mark for the entire assessment or part of it. If there is more than one guilty party, such as in case of collusion, all parties involved will receive the same penalty.

D. Instruction to Students

This is a group assignment for 5 students that will contribute 20% to your overall assessment. Each group must submit the following items via XMDrive (link provided on the Moodle system). Ensure the filenames follow the specified naming format: G0123_Assign1_Group1.pdf, G0123_Assign1_Group1.mp4, G0123_Assign1_Group1_1.m, G0123_Assign1_Group1.m:

1. **MATLAB Code (.m file)**: A well-commented script that performs the spinner simulation, visualizes the results, handles user input, and provides statistical analysis. Ensure the code is modular and easy to follow.
2. **Report (PDF file)**: A 5-10 pages document (excluding contribution section and appendix) explaining your methodology, code, results, and interpretation. Use Times New Roman, size 12, with 1.5 line spacing.
3. **Presentation (MP4 file)**: A 10-15 minutes video demonstrating your work, key findings, and visualizations.

Submission Deadline: Thursday, 06 March 2025 at 5:00 PM via XMDrive, the link is provided in Moodle system).

E. Evaluation Breakdown

No.	Component Title	Percentage (%)
1.	Spinner Design & Animation	25
2.	Simulation Accuracy	25
3.	Data Visualization	25
4.	Report & Presentation	25
	TOTAL	**100**

F. Task(s)

In this project, you will:

1. Design and animate a customizable spinner using MATLAB.
2. Simulate multiple spins and collect data on the frequency of each outcome.
3. Visualize the outcomes using various statistical plots (e.g., bar charts, pie charts, and cumulative frequency graphs).
4. Analyse the fairness of the spinner by comparing theoretical and empirical probabilities.
5. Document your work in a detailed report and present your findings through a recorded video presentation.

Project Requirements

1. Spinner Design and Customization

- Design a spinner wheel with at least 6 different sectors.
- Each sector should have a unique label, color, and probability weight.
- Implement a dynamic display that updates with each spin.

2. Simulate Multiple Spins

- Allow the user to input the number of spins (e.g., 500 spins).
- Simulate the spinner using random number generation.
- Record and store the outcome of each spin.

3. Visualize Simulation Results

- Spinner Animation: Display the spinner wheel during each spin (e.g., using a loop and updating the plot).
- Outcome Distribution:

    - Use a bar chart to show the frequency of each outcome.
    - Use a pie chart to visualize the proportion of outcomes.
- Cumulative Frequency: Plot a cumulative frequency graph that shows how outcomes evolve over multiple spins.
- Custom Visuals: Add at least one additional visualization (e.g., line plot of cumulative probabilities, a scatter plot of outcomes, or a custom spinner design visualization).

4. Probability Analysis

- Calculate the theoretical probability of each sector based on weights.
- Compare the empirical probability (from the simulation) with the theoretical values.
- Discuss whether the spinner is fair (i.e., do outcomes match their probabilities?).

Report Structure

1. **Introduction**: Explain the spinner game and its relation to probability theory.
2. **Methodology**: Describe how you designed the spinner, the random simulation process, and how you visualized the results.
3. **Results**: Include all key plots and explain the outcome patterns observed.
4. **Discussion**: Analyse whether the spinner is fair, identify any biases, and interpret your visualizations.
5. **Conclusion**: Summarize your key findings and reflect on the project's challenges and learning outcomes.
6. **Contribution of Group Members**: Clearly outline each member's specific contributions to the project (e.g., coding, visualization, report writing, presentation).
7. **Appendix**: Include your fully-commented MATLAB code(s).

Presentation Guidelines

1. Demonstrate the Spinner: Showcase your spinner simulation and animations.
2. Explain Key Plots: Walk through your statistical visualizations and interpret the results.
3. Analyse Fairness: Present your probability comparison and any observed biases.
4. Team Collaboration: Highlight contributions from each team member.

APPENDIX 1

MARKING RUBRICS

Component Title					**Percentage (%)**		
Criteria	**Score and Descriptors**					**Weight (%)**	**Marks**
	Excellent (5)	**Good (4)**	**Average (3)**	**Need Improvement (2)**	**Poor (1)**		
Spinner Design & Animation	Detailed, dynamic spinner with clear labels and colors	Functional spinner with some customization	Basic spinner with minimal customization	Incomplete spinner with poor visualization	No spinner or visualization	25%	
Simulation Accuracy	Accurate and robust simulation with user input handling	Correct simulation with minor inaccuracies	Functional simulation with limited input options	Incomplete simulation with errors	No functioning simulation	25%	
Data Visualization	Clear, accurate, and diverse plots (bar, pie, cumulative)	Clear and accurate plots with minor inconsistencies	Basic plots with limited detail	Poor visualization with missing or unclear plots	No visualization	25%	
Report & Presentation	Clear, well-organized, thorough, and engaging content	Well-organized with most required content	Includes basic information but lacks clarity	Disorganized, missing key sections	Missing or incomplete	25%	
						TOTAL	

Note to students: Please **DO NOT** include the marking rubric when submitting your coursework.